Shahzalal Khan Rahat

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Skills

ML	Supervised & Unsupervised Learning, AI
Data Analysis	Data Cleaning, Feature Engineering, Exploratory Data Analysis (EDA)
Computer Vision	Image Segmentation, Feature Extraction, Clustering
Soft Skills	Communication, Presentation, Stakeholder Collaboration

Tools

Tools	Python, Canva, LaTeX, MS Word, MS Excel

Education

2026 **B.Sc in Computer Science and Engineering**, *BRAC University*, Dhaka, Bangladesh. CGPA: 3.35

2020 **HSC**, *Cantonment Public School and College, Saidpur*, GPA: 5.00

2018 **SSC**, *Cantonment Public School and College, Saidpur*, GPA: 5.00

Research

Publication [Accepted] SEWING: An Interpretable Ensemble ML Framework for Sewing Operators' Wage Increments Prediction in the Garment Industry. Developed an interpretable ML framework analyzing 14 operational KPIs across 4 garment factories, achieving R^2 up to 0.99 to support data-driven wage and workforce decisions and improve operational efficiency in RMG production environments. DOI: https://doi.org/10.1016/j.eswa.2026.131900

[Accepted] LassoForest: CGPA Dynamics through Time-Series Forecasting and Counterfactual Analysis of Alcohol Consumption. This work suggest an temporal machine learning framework called LassoForest to forecast CGPA, which uses long term behavioral data such as alcohol consumption.

Thesis TorqueSAM: Unsupervised Kidney CT Analysis with Localization and SAM-Integrated Torque Clustering Segmentation Developed an unsupervised pipeline to segment renal stones using MedSAM and clustering as the core mechanism, avoiding supervised training labels.

Projects

The Travel Agency Platform Developed a ticket booking platform with hotel and weather history features, promo code support, and real-time data handling. **GitHub:** github.com/shahzalalrahat/The-Travel-Agency-Platform

EYE of Blind Built a sonar-based navigation aid using Arduino and ultrasonic sensors to help visually impaired individuals detect obstacles through vibration feedback.

Volunteer Work Experience

General member, Robotics Club of BRAC University (ROBU), IT Department

Certifications & Achievements

2017 Participant in National High School Programming.

2022 Completed Basics Of Robotics [BOR] From BRAC University.

2025 CVIS Training on Development of ML/DL Models by BRACu CVIS lab.

2025 Cleaning Data in Python. [Datacamp]

2025 Feature Engineering for Machine Learning in Python. [Datacamp]

Hobbies

Motorcycling, Cooking, watching tech reviews, watching and playing Cricket, Photography.
//drive.google.com/drive/folders/1BZ21$_cpHJOrmj8UmJRFS6lGSywKMXg34$

References

Thesis Supervisor
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